JungTao (Ronnie) Cheng

Ph.D. candidate in mathematics, Stanford University
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Education

Stanford University 2023–present
Ph.D. in Mathematics, in progress. Advisor: Ravi Vakil.

National Taiwan University 2018–2023
B.S. in Mathematics, with a double major in Computer Science.

Research Interests

Algebraic geometry, matroid theory, algebraic combinatorics, and algorithms, with particular interests in wonderful compactifications and their intersection theory.

Publications and Manuscripts

Publication

1. **On the tangent bundle and the divisor theory of a general matroid.**
   International Mathematics Research Notices **2026**, no. 17, rnag188.
   doi:10.1093/imrn/rnag188 | arXiv:2510.06609.

Preprints

2. **Kazhdan–Lusztig polynomials of matroids need not be unimodal.**
   With Shurui Liu.
   2026. arXiv:2607.24186.
3. **Tangent classes for matroid building sets.**
   2026. arXiv:2606.22650.
4. **Inequalities for Chow polynomials and Chern numbers of matroids.**
   With Wangyang Lin.
   2026. arXiv:2603.21680.

In Preparation

5. **Vanishing theorems for matroids and wonderful compactifications.**
   Manuscript in preparation, 2026. *Working title.*

AI for Mathematics

A1. **Danus: Orchestrating Mathematical Reasoning Agents with Fact-Graph Memory.**
With Jihao Liu, Guoxiong Gao, Zeming Sun, Bin Wu, Shurui Liu, Jiedong Jiang, Haocheng Ju, Leheng Chen, Xiping Zhang, and Bin Dong.
Preprint, 2026. arXiv:2607.06447.

Related experiment documentation: Tangent classes of matroids and wonderful compactifications, with Shurui Liu and Guoxiong Gao (2026), arXiv:2607.05835. Record of Danus-generated mathematical output; not intended for journal publication.

Research Talks and Presentations

University of California, Berkeley November 2026
Commutative Algebra and Algebraic Geometry Seminar. *Forthcoming.*

University of California, Berkeley October 2026
Combinatorics Seminar. *Forthcoming.*

Fields Institute / University of Toronto October 2026
Algebraic Combinatorics Seminar. *From tangent bundles of matroids to a stone game.*

Joint Mathematics Meetings January 2026
Special Session on Combinatorial Algebraic Geometry, Washington, D.C.

Western Algebraic Geometry Symposium (WAGS) November 2025
University of Oregon. Five-minute lightning talk.

Selected Honors and Fellowships

Olympiad Scholarship for Study Abroad 2023–2027
Ministry of Education, Taiwan.

Dean's Award, College of Science
National Taiwan University; recognition of overall undergraduate achievement.

ICPC World Finals 2021, 2022, 2024
National Taiwan University (18th and 27th); Stanford University (46th).

International Mathematical Olympiad 2016–2018
Gold medals in 2016 and 2018; silver medal in 2017. Best result: 6th place (2018).

Teaching

Stanford University. TA: Teaching Assistant; CA: Course Assistant.

Winter 2026	Math 51: Linear Algebra, Multivariable Calculus, and Modern Applications	TA
Fall 2025	Math 110: Number Theory for Cryptography	TA
Spring 2025	Math 120: Modern Algebra	TA
Fall 2024	Math 107: Graph Theory	CA
Spring 2024	Math 115: Functions of a Real Variable	CA
Fall 2023	Math 104: Applied Matrix Theory	CA

Mentoring and Outreach

Mentor, Directed Reading Program 2023–2024
Stanford University. Introduction to algebraic geometry; Fall 2023, Winter 2024, and Spring 2024.

Instructor, Mathematics Olympiad Training
Taiwan. IMO training camps and the High School Cultivate Math Talents Project.

Programming

Competitive programming. Codeforces: highest title International Grandmaster; peak rating 2625.